

Opgave 5

En ladning Q bevæges langs en cirkelbue med radius R og centrum i origo fra punktet $(R, 0, 0)$ til punktet $(\frac{\sqrt{3}}{2}R, \frac{1}{2}R, 0)$. Ladningen er i hvile både før og efter flytningen. I området er der et statisk elektrisk felt med formen

$$\mathbf{E} = -\frac{V_0}{R^3} (2xy \hat{\mathbf{x}} + x^2 \hat{\mathbf{y}}),$$

hvor V_0 er en konstant.

- a) Vis, at $\mathbf{E}$ beskriver et muligt elektrostatisk felt. Bestem ladningstætheden i området.

$$\vec{\nabla} \times \vec{E} = -\frac{V_0}{R^3} \left(\frac{\partial}{\partial x} x^2 - \frac{\partial}{\partial y} 2xy \right) \hat{\mathbf{z}} = 0 \quad \checkmark \quad \text{muligt elektrostatisk felt}$$

$$\rho = \epsilon_0 \vec{\nabla} \cdot \vec{E} = \epsilon_0 \frac{-V_0}{R^3} (2y + 0) = \underline{\underline{-\frac{2\epsilon_0 V_0}{R^3} y}}$$

- b) Bestem arbejdet udført mod feltet ved den beskrevne bevægelse af ladningen.

Ved integration af x - og y -komponenterne af feltet får vi

$$\frac{-V_0}{R^3} \int 2xy dx = \frac{-V_0}{R^3} x^2 y + C_1, \quad \frac{-V_0}{R^3} \int x^2 dy = \frac{-V_0}{R^3} x^2 y + C_2. \quad \text{Ud fra dette ses det at}$$

$V = \frac{V_0}{R^3} x^2 y + C$, hvor C er en integrationskonstant. Altså er arbejdet

$$W = Q \left[V\left(\frac{\sqrt{3}}{2}R, \frac{1}{2}R, 0\right) - V(R, 0, 0) \right] = \frac{QV_0}{R^3} \left(\left(\frac{\sqrt{3}}{2}R\right)^2 \frac{1}{2}R - R^2 \cdot 0 \right) = \frac{QV_0}{R^3} \frac{3}{4} R^2 \frac{1}{2}R = \underline{\underline{\frac{3}{8} QV_0}}$$

Alternativt: kurveintegrale $W = -Q \int_a^b \vec{E} \cdot d\vec{l}, \quad (R, 0, 0) \rightarrow \left(\frac{\sqrt{3}}{2}R, 0, 0\right) \rightarrow \left(\frac{\sqrt{3}}{2}R, \frac{1}{2}R, 0\right)$

$$(i): E_x = 0 \rightarrow W = 0. \quad (ii): x = \frac{\sqrt{3}}{2}R, \quad d\vec{l} = dy \hat{\mathbf{y}}, \quad y: 0 \rightarrow \frac{1}{2}R, \quad E_y = \frac{-V_0}{R^3} \left(\frac{\sqrt{3}}{2}R\right)^2 = \frac{3}{4} \frac{-V_0}{R}$$

$$W = -Q \int_0^{R/2} \frac{3}{4} \frac{-V_0}{R} dy = \frac{3}{4} \frac{QV_0}{R} \frac{R}{2} = \underline{\underline{\frac{3}{8} QV_0}}$$

Problem 2.50 Suppose an electric field $\mathbf{E}(x, y, z)$ has the form

$$E_x = ax, \quad E_y = 0, \quad E_z = 0$$

where a is a constant. What is the charge density? How do you account for the fact that the field points in a particular direction, when the charge density is uniform? [This is a more subtle problem than it looks, and worthy of careful thought.]

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E} = \underline{\epsilon_0 a}$$

Føltet $\mathbf{E} = \frac{a}{3}(x\hat{x} + y\hat{y} + z\hat{z})$ eller $\mathbf{E} = ay\hat{y}$ giver samme ρ

Ud fra Gauss' lov. Et givent felt kan ikke defineres udelukkende

ved $\rho = \epsilon_0 \nabla \cdot \mathbf{E}$ eller $\nabla \times \mathbf{E}$. Det kræver de rette grænsebetingelser

f.eks. $\mathbf{E} \rightarrow 0$, når $r \rightarrow \infty$ eller symmetriargumenter.

I dette tilfælde har vi ingen af ovenstående og vi kan dermed ikke finde $\mathbf{E}$ -feltet fra en ladningsdensitet, der er konstant og uniform overalt.

Problem 2.32 Find the energy stored in a uniformly charged solid sphere of radius R and charge q . Do it three different ways:

(a) Use Eq. 2.43. You found the potential in Prob. 2.21.

(b) Use Eq. 2.45. Don't forget to integrate over *all space*.

$$a) W = \frac{1}{2} \int \rho V d\tau. \quad \rho = \frac{q}{\frac{4}{3}\pi R^3} = \frac{3q}{4\pi R^3}. \quad V = \frac{\rho}{2\epsilon_0} \left(R^2 - \frac{r^2}{3} \right) = \frac{3q}{8\pi\epsilon_0 R^3} \left(R^2 - \frac{r^2}{3} \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{2R} \left(3 - \frac{r^2}{R^2} \right)$$

$$W = \frac{1}{2} \int_0^{2\pi} \int_0^\pi \int_0^R \frac{3q}{4\pi R^3} \frac{1}{4\pi\epsilon_0} \frac{q}{2R} \left(3 - \frac{r^2}{R^2} \right) r^2 \sin\theta dr d\theta d\phi = \frac{1}{4\pi\epsilon_0} \frac{3q^2}{16\pi R^4} 4\pi \int_0^R \left(3r^2 - \frac{r^4}{R^2} \right) dr =$$

$$\frac{1}{4\pi\epsilon_0} \frac{3q^2}{4R^4} \left[r^3 - \frac{r^5}{5R^2} \right]_0^R = \frac{1}{4\pi\epsilon_0} \frac{3q^2}{4R^4} \frac{4}{5} R^3 = \underline{\frac{1}{4\pi\epsilon_0} \frac{3q^2}{5R}}$$

$$b) W = \frac{\epsilon_0}{2} \int E^2 d\tau, \quad r > R: \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}, \quad r \leq R: \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r \hat{r}$$

$$W = \frac{\epsilon_0}{2} \int_0^{2\pi} \int_0^\pi \int_0^\infty E^2 r^2 \sin\theta dr d\theta d\phi = \frac{4\pi\epsilon_0}{2} \left[\int_0^R \left(\frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r \right)^2 r^2 dr + \int_R^\infty \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right)^2 r^2 dr \right] =$$

$$\frac{1}{4\pi\epsilon_0} \frac{q^2}{2} \left(\left[\frac{r^5}{5R^6} \right]_0^R + \left[\frac{-1}{r} \right]_R^\infty \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{2} \left(\frac{1}{5R} - \frac{1}{\infty} + \frac{1}{R} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{2} \frac{6}{5R} = \underline{\frac{1}{4\pi\epsilon_0} \frac{3q^2}{5R}}$$

Sphere $\rightarrow$ hollow
ball $\rightarrow$ solid

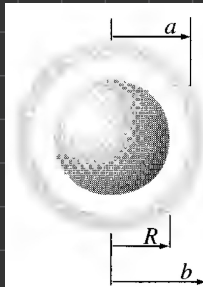


Figure 2.48

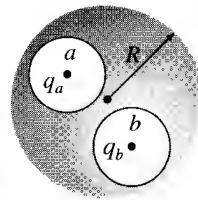


Figure 2.49

Problem 2.35 A metal sphere of radius R , carrying charge q , is surrounded by a thick concentric metal shell (inner radius a , outer radius b , as in Fig. 2.48). The shell carries no net charge.

- Find the surface charge density σ at R , at a , and at b .
- Find the potential at the center, using infinity as the reference point.
- Now the outer surface is touched to a grounding wire, which lowers its potential to zero (same as at infinity). How do your answers to (a) and (b) change?

$$a) \sigma_R = \frac{q}{4\pi R^2}, \sigma_a = \frac{-q}{4\pi a^2}, \sigma_b = \frac{q}{4\pi b^2}$$

$$b) V(0) = -\int_{\infty}^0 \vec{E} \cdot d\vec{l} = -\int_{\infty}^b \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr - \int_b^a 0 dr - \int_a^R \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr - \int_R^0 0 dr = \frac{1}{4\pi\epsilon_0} \left(\left[\frac{q}{r} \right]_{\infty}^b + \left[\frac{q}{r} \right]_a^R \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{b} + \frac{q}{R} - \frac{q}{a} \right)$$

$$c) \text{ Skallen bliver afladt så } \sigma_b \rightarrow 0. V(0) = -\int_{\infty}^a 0 dr - \int_a^R \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr - \int_R^0 0 dr = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R} - \frac{q}{a} \right)$$

Problem 2.36 Two spherical cavities, of radii a and b , are hollowed out from the interior of a (neutral) conducting sphere of radius R (Fig. 2.49). At the center of each cavity a point charge is placed—call these charges q_a and q_b .

- Find the surface charges σ_a , σ_b , and σ_R .
- What is the field outside the conductor?
- What is the field within each cavity?
- What is the force on q_a and q_b ?
- Which of these answers would change if a third charge, q_c , were brought near the conductor?

$$a) \sigma_a = \frac{-q_a}{4\pi a^2}, \sigma_b = \frac{-q_b}{4\pi b^2}, \sigma_R = \frac{q_a + q_b}{4\pi R^2}$$

$$b) \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q_a + q_b}{r^2} \hat{r}, \text{ hvor } \vec{r} \text{ er vektoren fra centrum af sfæren}$$

$$c) \vec{E}_a = \frac{1}{4\pi\epsilon_0} \frac{q_a}{r_a^2} \hat{r}_a, \vec{E}_b = \frac{1}{4\pi\epsilon_0} \frac{q_b}{r_b^2} \hat{r}_b, \text{ hvor } \vec{r}_a (\vec{r}_b) \text{ er vektoren fra centrum af hulrummet}$$

d) $\vec{E}_a = \vec{E}_b = 0$, bortset fra i hulrummene er $\vec{E} = 0$ overalt i lederen. Ladningen kan ikke udøve en kraft på sig selv.

e) σ_R og $\vec{E}$ ændrer sig. q_c forskyder ladningerne på overfladen og gør σ_R ikke-uniform. $\vec{E}$ afhænger nu også af q_c og dens lokation.

Problem 2.39 Find the capacitance per unit length of two coaxial metal cylindrical tubes, of radii a and b (Fig. 2.53).

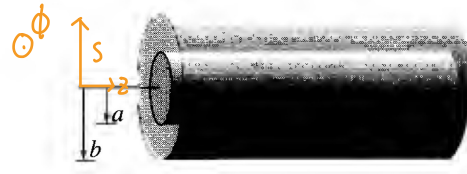


Figure 2.53

Vi antager at den indre cylinder har ladningen Q for en given længde L

Følgelig fås da ved Gauss' lov $\oint \vec{E} \cdot d\vec{a} = E 2\pi s L = \frac{Q_{enc}}{\epsilon_0} = \frac{Q}{L} \Rightarrow \vec{E}(s) = \frac{Q}{2\pi\epsilon_0 L} \frac{1}{s} \vec{s}, a \leq s \leq b$

$$V(b) - V(a) = - \int_a^b \vec{E} \cdot d\vec{l} = - \frac{Q}{2\pi\epsilon_0 L} \int_a^b \frac{1}{s} ds = - \frac{Q}{2\pi\epsilon_0 L} [\ln s]_a^b = - \frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{b}{a}\right)$$

Potentialforskellen $V = V_+ - V_- = V(a) - V(b) = -(V(b) - V(a)) = \frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{b}{a}\right)$

$$C = \frac{Q}{V} = \frac{2\pi\epsilon_0 L}{\ln(b/a)}, \text{ pr. længdeenhed fås } \underline{C = \frac{2\pi\epsilon_0}{\ln(b/a)}}$$

Opgave 5

Vi betragter en kubisk område i rummet, $\mathcal{V}$, defineret ved $0 < x < L$, $0 < y < L$ og $0 < z < L$, hvor L angiver en længde.

I $\mathcal{V}$ er et statisk vektorfelt givet ved

$$\mathbf{f}(x, y, z) = C \cdot [y^2 \hat{\mathbf{x}} + (2xy + z^2) \hat{\mathbf{y}} + 2yz \hat{\mathbf{z}}],$$

hvor C er en konstant.

- a) Vis, at $\mathbf{f}$ opfylder det grundlæggende krav til et statisk elektrisk felt, men *ikke* til et magnetisk felt.

For et elektrostatisk felt gælder $\bar{\nabla} \times \bar{\mathbf{E}} = 0$, for et magnetisk $\bar{\nabla} \cdot \bar{\mathbf{B}} = 0$

$$\bar{\nabla} \times \bar{\mathbf{f}} = C [(2z - 2z) \hat{\mathbf{x}} + (0 - 0) \hat{\mathbf{y}} + (2y - 2y) \hat{\mathbf{z}}] = \bar{0} \quad \checkmark$$

$$\bar{\nabla} \cdot \bar{\mathbf{f}} = C (0 + 2x + 2y) = 2C(x + y) \neq 0 \quad \times$$

- b) Angiv enheden for C , således at $\mathbf{f}$ beskriver et elektrisk felt, $\mathbf{E}(x, y, z) = \mathbf{f}(x, y, z)$.

$$[E] = \frac{N}{C} = \frac{V}{m} \cdot [f] = [C] m^2 = \frac{V}{m} \Leftrightarrow [C] = \frac{V}{m^3}$$

- c) Bestem den rumlige ladningstæthed, $\rho(x, y, z)$, konsistent med feltet $\mathbf{E}(x, y, z)$.
Bestem dernæst den totale ladning i $\mathcal{V}$.

$$\bar{\rho} = \epsilon_0 \bar{\nabla} \cdot \bar{\mathbf{E}} = 2\epsilon_0 C(x + y)$$

$$Q = \int \rho d\tau = 2\epsilon_0 C \int_0^L \int_0^L \int_0^L (x + y) dx dy dz = 2\epsilon_0 C L^2 \left(\frac{L^2}{2} + \frac{L^2}{2} \right) = 2\epsilon_0 C L^4$$

Problem 3.2 In one sentence, justify **Earnshaw's Theorem**: *A charged particle cannot be held in a stable equilibrium by electrostatic forces alone.* As an example, consider the cubical arrangement of fixed charges in Fig. 3.4. It looks, off hand, as though a positive charge at the center would be suspended in midair, since it is repelled away from each corner. Where is the leak in this "electrostatic bottle"? [To harness nuclear fusion as a practical energy source it is necessary to heat a plasma (soup of charges particles) to fantastic temperatures—so hot that contact would vaporize any ordinary pot. Earnshaw's theorem says that electrostatic containment is also out of the question. Fortunately, it is possible to confine a hot plasma magnetically.]

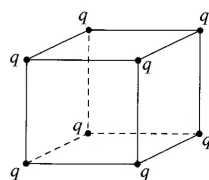
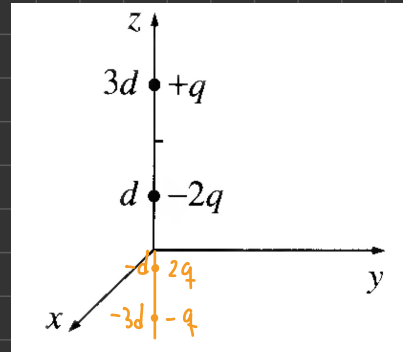


Figure 3.4

En stabil ligevægt er et lokalt minimum i den potentielle energi, som for denne opstilling er qv. Men fra Laplace' ligning ved vi, at der ikke eksisterer lokale minima. Derfor må det, der i opstillingen ligner et minimum, være et saddepunkt.

Problem 3.6 Find the force on the charge $+q$ in Fig. 3.14. (The xy plane is a grounded conductor.)



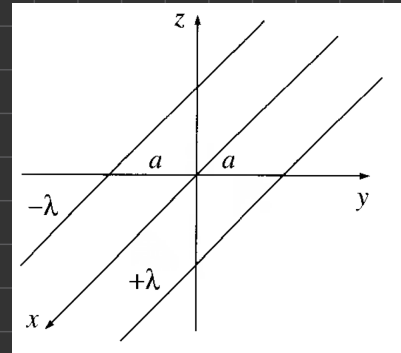
Vi indtegner billedeladningerne $2q$ ved $z=-d$ og $-q$ ved $z=-3d$. Kraften er da $\vec{F} = \frac{q}{4\pi\epsilon_0} \left(\frac{-2q}{(2d)^2} + \frac{2q}{(4d)^2} + \frac{-q}{(6d)^2} \right) \hat{z} =$

$$\frac{1}{4\pi\epsilon_0} \frac{q^2}{d^2} \left(\frac{-1}{2} + \frac{1}{8} - \frac{1}{36} \right) \hat{z} = \underline{\underline{\frac{-1}{4\pi\epsilon_0} \frac{29q^2}{72d^2} \hat{z}}}$$

Problem 2.47 Two infinitely long wires running parallel to the x axis carry uniform charge densities $+\lambda$ and $-\lambda$ (Fig. 2.54).

(a) Find the potential at any point (x, y, z) , using the origin as your reference.

Fra Prob. (2.23) $V = \frac{-\lambda}{2\pi\epsilon_0} \ln\left(\frac{s}{a}\right)$, hvor s er afstanden fra ledningen

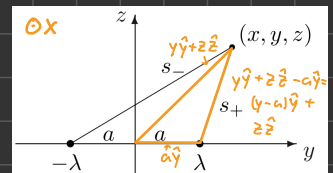


Fra λ_+ fås $V_+ = \frac{-\lambda}{2\pi\epsilon_0} \ln\left(\frac{s_+}{a}\right)$, hvor s_+ er afstanden fra λ_+
 $\lambda_- \Rightarrow V_- = \frac{-(-\lambda)}{2\pi\epsilon_0} \ln\left(\frac{s_-}{a}\right) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{s_-}{a}\right)$

$$V = V_+ + V_- = \frac{-\lambda}{2\pi\epsilon_0} \ln\left(\frac{s_+}{a}\right) + \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{s_-}{a}\right) = \frac{\lambda}{2\pi\epsilon_0} (-\ln s_+ + \ln a + \ln s_- - \ln a) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{s_-}{s_+}\right)$$

$$s_+ = \sqrt{(y-a)^2 + z^2}, \quad s_- = \sqrt{(y+a)^2 + z^2}$$

$$V(x, y, z) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{\sqrt{(y+a)^2 + z^2}}{\sqrt{(y-a)^2 + z^2}}\right) = \frac{\lambda}{4\pi\epsilon_0} \ln\left(\frac{(y+a)^2 + z^2}{(y-a)^2 + z^2}\right)$$

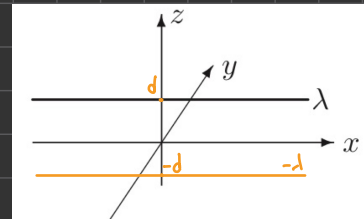


Problem 3.9 A uniform line charge λ is placed on an infinite straight wire, a distance d above a grounded conducting plane. (Let's say the wire runs parallel to the x -axis and directly above it, and the conducting plane is the xy plane.)

(a) Find the potential in the region above the plane.

(b) Find the charge density σ induced on the conducting plane.

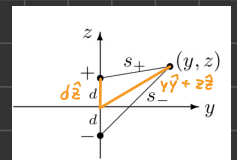
a) Billede opgave med ledning λ ved $z=d$, og ledning



$-\lambda$ ved $z=-d$. Vi kender potentialet fra Prob. 2.53

$$V(y, z) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{s_-}{s_+}\right) = \frac{2\lambda}{4\pi\epsilon_0} \ln\left(\frac{s_-}{s_+}\right) = \frac{\lambda}{4\pi\epsilon_0} \ln\left(\frac{s_-^2}{s_+^2}\right), \quad s_-^2 = y^2 + (z+d)^2,$$

$$s_+^2 = y^2 + (z-d)^2 \Rightarrow \underline{\underline{V(y, z) = \frac{\lambda}{4\pi\epsilon_0} \ln\left(\frac{y^2 + (z+d)^2}{y^2 + (z-d)^2}\right)}}$$



$$b) \sigma = -\epsilon_0 \frac{\partial V}{\partial n}, \quad \frac{\partial V}{\partial n} = \frac{\partial V}{\partial z} \Big|_{z=0}$$

$$\sigma = -\epsilon_0 \frac{\lambda}{4\pi\epsilon_0} \left(\frac{2z+2d}{y^2+(z+d)^2} - \frac{2z-2d}{y^2+(z-d)^2} \right) \Big|_{z=0} = \frac{-\lambda}{4\pi} \left(\frac{2d}{y^2+d^2} + \frac{2d}{y^2+d^2} \right) = \frac{-\lambda d}{\pi(y^2+d^2)}$$

Problem 2.37 Two large metal plates (each of area A) are held a distance d apart. Suppose we put a charge Q on each plate; what is the electrostatic pressure on the plates?

$E=0$ mellem pladerne og $\vec{E} = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0} \hat{n}$ uden for pladerne

$$(2.52) \quad P = \frac{\epsilon_0}{2} E^2 = \frac{\epsilon_0}{2} \frac{Q^2}{A^2 \epsilon_0^2} = \frac{Q^2}{2A^2 \epsilon_0}$$

Problem 2.38 A metal sphere of radius R carries a total charge Q . What is the force of repulsion between the "northern" hemisphere and the "southern" hemisphere?

$$(2.51) \quad \vec{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{n}, \quad \sigma = \frac{Q}{4\pi R^2}, \quad \hat{n} = \vec{r}, \quad f_z = \vec{f} \cos \theta, \quad F_z = \int_S f_z da$$

$$f_z = \frac{1}{2\epsilon_0} \left(\frac{Q}{4\pi R^2} \right)^2 \cos \theta, \quad F_z = \frac{1}{2\epsilon_0} \left(\frac{Q}{4\pi R^2} \right)^2 \int_0^{2\pi} \int_0^{\pi/2} R^2 \cos \theta \sin \theta d\theta d\phi =$$

$$\frac{1}{2\epsilon_0} \left(\frac{Q}{4\pi R^2} \right)^2 2\pi R^2 \left[\frac{1}{2} \sin^2 \theta \right]_0^{\pi/2} = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{8R^2}$$

